READING

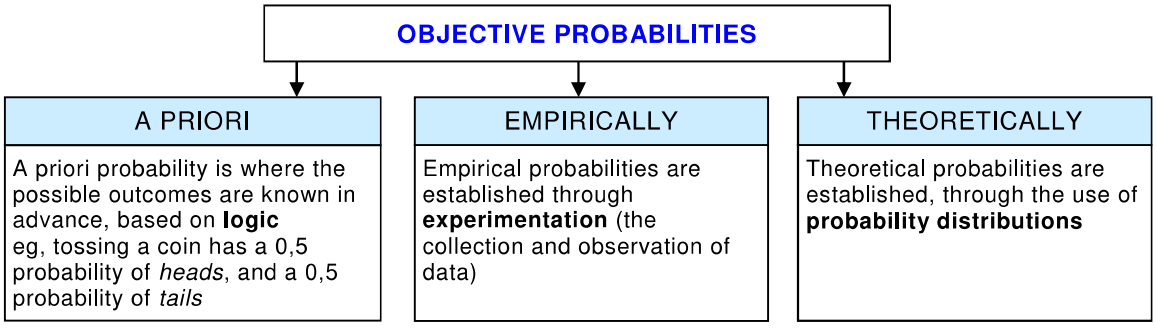
Prescribed textbook: Chapter 4

4.1 Introduction

This unit introduces the basic concepts of probability. It outlines rules and techniques for assigning probabilities to events. Probability plays a critical role in statistics. All of us form simple probability conclusions in our daily lives. Sometimes these determinations are based on fact, while others are subjective. If the probability of an event is high, one expects the event to occur rather than not to occur. If the probability of rain is more than 50%, it is more likely to rain than not.

4.2 Establishing probabilities

Probabilities can be based on educated guesswork (subjective probabilities) or they can be objectively established.



Definition: Probability:

Probability can be defined as a measure of ***how likely*** it is that an event will occur.

The probability of an event occurring is ***represented by a number that lies between 0 (impossible) and 1 (certain)***.

Probabilities are expressed as proportions or factors, eg 0,5, rather than as percentages, eg 50%, although in speaking, we tend to use percentages.

If you conduct a trial, eg toss a coin, there will be an outcome (either a head or a tail). Because there exists only 2 possible outcomes in your trial, there is an even chance (0,5 probability) of the outcome being heads, and a 0,5 probability of the outcome being tails with all possible outcomes adding up to 1.

4.3 Probability symbols

Commonly used symbols when describing probability relationships are:

Symbol	Meaning
$P(...)$	Probability of whatever event is stated between brackets.
$P(A)$	Probability of an event of a specific type (or specific properties).
$P(\bar{A})$	Probability that the event will not occur.
$P(A \cap B)$	The symbol is read as 'and'; this is the probability that event A and B will occur.
$P(A \cup B)$	The symbol is read as 'or'; this is the probability that event A will occur or event B will occur.
$P(A\|B)$	Probability of A, given that B happens/does something.

4.4 Properties of a probability

Five basic probability requirements apply:

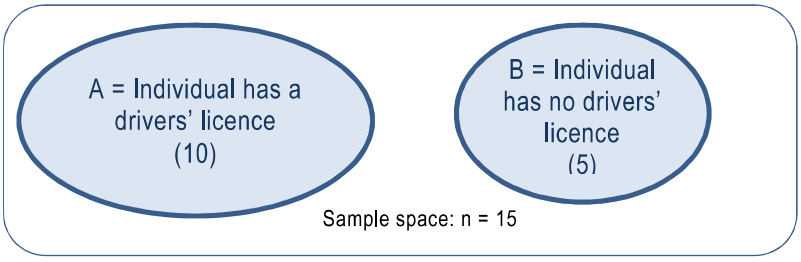
1.	Each even probability lies between 0 and 1 inclusive.
2.	If an event cannot occur, then $P(A) = 0$
3.	If an event is certain to occur, then $P(A) = 1$
4.	The sum of the probabilities of all possible events in the sample space equals 1, ie $P(A_1) + P(A_2) + P(A_3) + P(A_4) + \ldots P(A_n) = 1$ for n events.
5.	If $P(A)$is the probability of an event occurring, then the probability of the event not occurring is $P(\bar{A}) = 1 - P(A)$

4.5 Probability concepts

Events are said to be ***mutually exclusive*** if they cannot occur together in a single trial, ie at the same time.

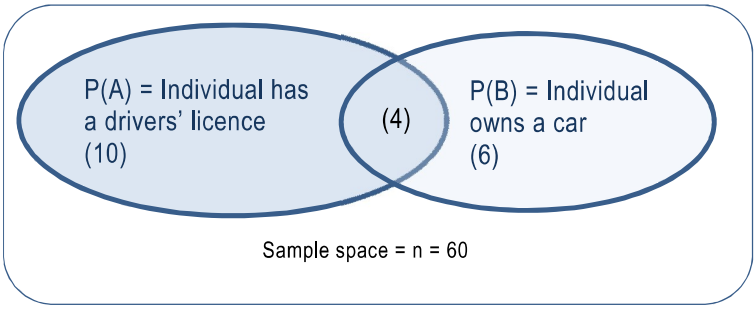
EXAMPLE

An individual either has a drivers' licence or he does not.



These events ***are mutually exclusive*** as there is no possibility that an individual both has a drivers' licence and does not have one (there is no overlap in the Venn diagrams).

However, an individual can both own a car and have a drivers' licence:



In this case the events ***are not mutually exclusive***.

4.6 Calculating probabilities

Definition: Marginal probability $P(A)$

A **marginal probability** is the probability of a single event occurring.

Marginal probability is denoted by $P(A)$.

EXAMPLE

A sample of university students indicates degree courses studied by gender:

Course	Gender		Total
	Male	Female	
BCom	20	18	38
BA	14	18	32
BSc	23	27	50
Total	57	63	120

Note: this table is known as a ***contingency table***.

What is the probability of a student being female?

$$P(female) = \frac{63}{120} = 0{,}525 \; or \; 52{,}5\%$$

There is a 0,525 probability or a 52,5% chance that a student is female.



TIP

It is recommended that you round all probability calculations to 4 decimal places. That way if you do need to present your results as a percentage, the percentage value will be to 2 decimal places.

Definition: Joint probability $P(A \cap B)$

Joint probability is the probability that event A and event B will occur at the same time in a single trial.

Joint probability is denoted by$P(A \cap B)$.

Example

Using the table in the previous example, calculate the probability that a BCom student will be male.

$$P(BCom \cap male) = \frac{20}{120} = 0{,}1667 \; or \; 16{,}67\%$$

There is a 0,1667 probability or 16,67% chance that a BCom student twill be male.

If events are mutually exclusive, there is no joint probability.



TIP

In sense-testing answers to probability questions, remember, probability can never be greater than 1.

Definition: Conditional probability $P(A|B)$

Conditional probability is the probability that event A will occur given that event B has already occurred.

Conditional probability is denoted by$P(A|B)$.

Formula: Conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Important point about conditional probability: when probability is conditional, the sample space is reduced to only those outcomes associated with the given event.



TIP

The key to recognising conditional probability is to look for the words "given that" or “knowing that” or their equivalent.

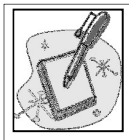
Example

Using the table in the previous example, calculate the probability that a student is studying BCom given that the student is male.

Instead of dividing by the total sample space of 120, the division is by only the male students, ie the sample space has been reduced to only males, ie 57 students.

$$P(BCom|male) = \frac{P(BCom \cap male)}{P(male)} = \frac{20}{57} = 0{,}3509 \; or \; 35{,}09\%$$

There is a 0,3509 probability or 35,09% chance that a student is studying BCom given that the student is male.



SELF-ASSESSMENT ACTIVITY

A company's employees are classified according to age and salary:

Age	Salary			Total
	< $25 000	$25 000 - $45 000	> $45 000	
< 30	32	3	0	35
30 – 45	10	18	21	49
> 45	1	10	5	16
Total	43	31	26	100

One employee is selected at random and two events are defined as:

A The employee is under 30

B The employee's salary is below $25 000.

Calculate each probability and express in words:

a) $P(A|B)$

b) $P(B|A)$

c) $P(A|\bar{B})$

d) $P(\bar{A}|B)$

SOLUTION TO SELF-ASSESSMENT ACTIVITY

a) Given that the employee's salary is below $25 000, the probability that the employee is under 30:

$$P(A|B) = \frac{P(A\ and\ B)}{P(B)} = \frac{\frac{32}{100}}{\frac{32+10+1}{100}} = \frac{32}{43} = 0{,}7442\ or\ 74{,}42\%$$

Alternatively, rather than using the formula directly, we can find $P(A|B)$ by simply consulting the contingency table. Given that the selected employee's salary is below $25 000, the sample space of possible outcomes is reduced to the 43 employees who fall into this category. Of these employees, 32 are under the age of 30.

$$P(A|B) = \frac{32}{43} = 0{,}7442\ or\ 74{,}42\%$$

b) Given that the employee is under 30, the probability that the employee's salary is below $25 000:

$$P(B|A) = \frac{P(A\ and\ B)}{P(A)} = \frac{\frac{32}{100}}{\frac{32+3+1}{100}} = \frac{32}{35} = 0{,}9143\ or\ 91{,}43\%$$

Or directly from the table, by reducing the sample size to the 35 under age 30 employees.

c) Given that the employee's salary is at least \$25 000, the probability that the employee is under 30:

$$P(A|\bar{B}) = \frac{P(A \text{ and } \bar{B})}{P(\bar{B})} = \frac{\frac{3+0}{100}}{\frac{31+26}{100}} = \frac{3}{57} = 0{,}0526 \text{ or } 5{,}26\%$$

Or directly from the table, by reducing the sample size to the 57 employees with salaries above \$25 000 (31+26).

d) Given that the employee's salary is below \$25 000, the probability that the employee is at least 30:

$$P(\bar{A}|B) = \frac{P(\bar{A} \text{ and } B)}{P(B)} = \frac{\frac{10+1}{100}}{\frac{43}{100}} = \frac{11}{43} = 0{,}2558 \text{ or } 25{,}58\%$$

Or directly from the table, by reducing the sample size to the 43 employees with salaries below \$25 000.

4.7 Probability rules

4.7.1 Addition rule

The addition rule is used to find the probability of either of one or more events occurring. It includes the probability that events occur simultaneously if they are mutually exclusive events.

Formula/rule: Addition rule for non-mutually exclusive events

The probability of either event A occurring or event B occurring (or more) or both events occurring in a single trial:

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

The reason for deducting the joint probability is to ensure the 'overlap' is not counted twice.



TIP

The 'or' in probability denotes addition, or +.

EXAMPLE

Using the student sample in the previous example, calculate the probability that the student is male or is studying BCom or both.

$$P(male \cup BCom) = P(male) + P(BCom) - P(male \cap BCom) = \frac{57}{120} + \frac{38}{120} - \frac{20}{120} = \frac{75}{120}$$

$$= 0{,}625 \text{ or } 62{,}5\%$$

There is a 0,625 probability or 62,5% chance that a student is male or is studying BCom or both.

Formula/rule: Addition rule for mutually exclusive events

The probability of either event A occurring or event B occurring in a single trial (by definition both events cannot occur simultaneously):

$P(A \cup B) = P(A) + P(B)$

EXAMPLE

Using the student sample in the previous example, calculate the probability that the student is studying BSc or studying BCom.

$$P(BSc \cup BCom) = P(BSc) + P(BCom) = \frac{50}{120} + \frac{38}{120} = \frac{88}{120} = 0{,}7333\ or\ 73{,}33\%$$

There is a 0,7333 probability or 73,33% chance that a student is studying BSc or BCom.

EXAMPLE

The distribution of blood types in a certain country is roughly:

A: 41% B: 9% AB: 4% O: 46%

An individual is brought into the emergency room after an automobile accident. What is the probability that he will be of type A or B or AB?

$$P(A\ or\ B\ or\ AB) = P(A) + P(B) + P(AB) = 0{,}41 + 0{,}09 + 0{,}04 = 0{,}554$$

Since it is impossible for one individual to have two different blood types, these events are mutually exclusive.

4.7.2 Multiplication rule

The multiplication rule is used to establish the joint probability of two events occurring at the same time in a single trial.

Formula/rule: Multiplication rule for statistically dependent events

The probability of both event A occurring and event B occurring in a single trial. The events are associated (ie, they are statistically dependent).

$P(A \cap B) = P(A|B) \times P(B)$

EXAMPLE

Using the student sample in the previous example, calculate the probability that the student is studying BCom and is male.

$$P(BCom \cup male) = P(BCom|male) \times P(male) = \frac{20}{57} \times \frac{57}{120} = \frac{20}{120} = 1{,}667\ or\ 16{,}67\%$$

There is a 0,1667 probability or 16,67% chance that a student is studying BCom and is male.

(You'll note this is the same as the answer in example, which proves that the events are dependent).

Formula/rule: Multiplication rule for statistically independent events

The probability of both event A occurring and event B occurring in a single trial. The events are independent.

$P(A \cap B) = P(A) \times P(B)$



TIP

The 'and' in probability denotes multiplication or X

EXAMPLE

In a pack of 52 cards, there are 13 cards in each of the four suits, spades, hearts, diamonds and clubs. There are 4 kings in the pack, one of each suit.

What is the probability of drawing a king of diamonds from the pack?

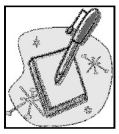
$$P(king \cup diamond) = P(king) \times P(diamond) = \frac{4}{52} \times \frac{13}{52} = \frac{1}{13} \times \frac{13}{52} = \frac{1}{52} = 0{,}0192 \; or \; 1{,}92\%$$

There is a 0,0192 probability or 1,92% chance that a card drawn from the pack is the king of diamonds.

EXAMPLE

The probability that a certain plant will flower during the first summer is 0,6. If five plants are planted, calculate the probability that all of them will have flowers during the first summer.

$$P(all\ flowers) = P(1st\ plant\ flowering\ and\ 2nd\ and\ 3rd\ and\ 4th\ and\ 5th\ and\ 6th)$$
$$= P(1st) \times P(2nd) \times P(3rd) \times P(4th) \times P(5th) \times P(6)$$
$$= 0{,}6 \times 0{,}6 \times 0{,}6 \times 0{,}6 \times 0{,}6 \times 0{,}6 = 0{,}078$$



SELF-ASSESSMENT ACTIVITY

A manufacturing plant conducted a survey to determine its employees' reactions toward a proposed change in working hours. A breakdown of the responses is:

Division	Agree	Disagree
Production	17	23
Office	8	2

Suppose an employee is chosen at random, with the relevant events being defined as:

A: The employee works in production.

B: The employee agrees with the proposed change.

Express each of the following events in words:

a) $\bar{A}$

b) $(A\ or\ B)$

c) $(A\ and\ B)$

d) $(A\ or\ \bar{B})$

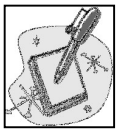
SOLUTION TO SELF-ASSESSMENT ACTIVITY

a) $\bar{A}$ The employee works in the office.

b) $(A\ or\ B)$ The employee either works in production or agrees with the proposed change or both.

c) $(A\ and\ B)$ The employee works in production and agrees with the proposed change.

d) $(A\ or\ \bar{B})$ The employee either works in production or disagrees with the proposed change or both.



SELF-ASSESSMENT ACTIVITY

Refer to the previous self-assessment activity and find:

a) $P(\bar{A})$

b) $P(A\ or\ B)$

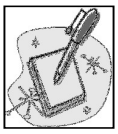
c) $P(A\ and\ B)$

SOLUTION TO SELF-ASSESSMENT ACTIVITY

a) $P(\bar{A}) = \frac{8}{50} + \frac{2}{50} = \frac{10}{50} = 0{,}2$

b) $P(A\ or\ B) = \frac{17}{50} + \frac{23}{50} = \frac{48}{50} = 0{,}96$

c) $P(A\ and\ B) = \frac{17}{50} = 0{,}34$



SELF-ASSESSMENT ACTIVITY

Wild azaleas are classified by colour and by the presence and absence of fragrance.

Fragrance	White	Pink	Orange	Total
Yes	12	60	58	130
No	50	10	10	70
Total	62	70	68	200

If an azalea is randomly selected from the group, calculate:

a) $P(fragrance)$

b) $P(orange)$

c) $P(orange\ with\ fragrance)$

d) $P(orange\ knowing\ it\ has\ fragrance)$

e) $P(has\ fragrance\ given\ that\ it\ is\ orange)$

SOLUTION TO SELF-ASSESSMENT ACTIVITY

a) $P(fragrance) = \frac{130}{200} = 0{,}65$

b) $P(orange) = \frac{68}{200} = 0{,}34$

c) $P(orange\ with\ fragrance) = \frac{58}{200} = 0{,}29$

d) $P(orange\ knowing\ it\ has\ fragrance) = \frac{58}{130} = 0{,}4462$

e) $P(has\ fragrance\ given\ that\ it\ is\ orange) = \frac{58}{68} = 0{,}8529$